COMPOSING PSEUDO-COHERENT MORPHISMS OF SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 17401-17405 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 17407-17416, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. BORDERLINE: pointer proof ('This follows from either ... Lemma ... or ... Lemma ...') with two exact labels; flagged for the Chief.

Editorial note. Counted A side only (spaces twin proof incomplete per ruling v3). Workhorse cited by L16234's proof.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. More on Algebra, lemmas pull-relative-pseudo-coherent, composition-relative-pseudo-coherent.

Source reference. Original lemma 067D: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17416–17416. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 067B: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 17416–17416. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *A composition of pseudo-coherent morphisms of schemes is pseudo-coherent.*

Proof. This translates into the following algebra result: If $A \to B \to C$ are composable pseudo-coherent ring maps then $A \to C$ is pseudo-coherent. This follows from either More on Algebra, Lemma 067B or More on Algebra, Lemma 067D. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.